Conic Sections

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Degrees, radians, and angles as rotations

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

2 [Foundational] • Target: Degrees, radians, and angles as rotations

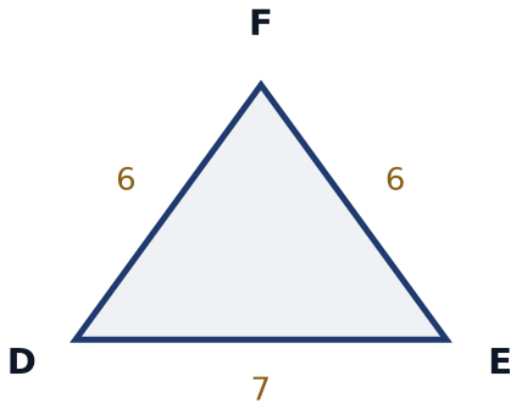
On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

3 [Foundational] • Target: Degrees, radians, and angles as rotations

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

4 [Foundational] • Target: Degrees, radians, and angles as rotations

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.



5 [Foundational] • Target: Degrees, radians, and angles as rotations

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

6 [Foundational] • Target: Degrees, radians, and angles as rotations

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Degrees, radians, and angles as rotations

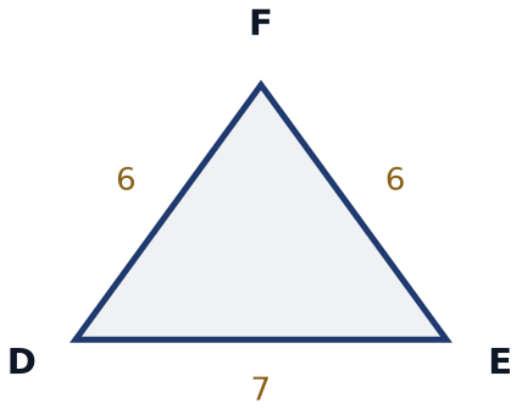
A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

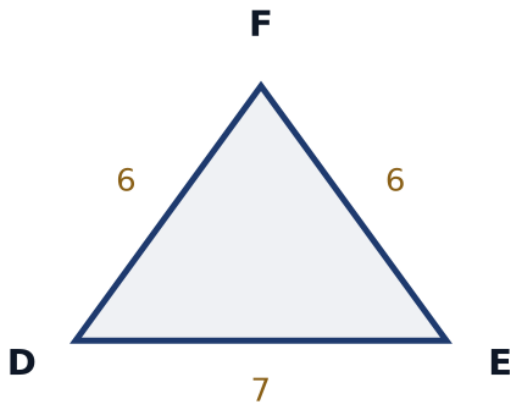
8 [Intermediate] • Target: Degrees, radians, and angles as rotations

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



9 [Intermediate] • Target: Degrees, radians, and angles as rotations

In ΔABC studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 [Intermediate] • Target: Degrees, radians, and angles as rotations

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

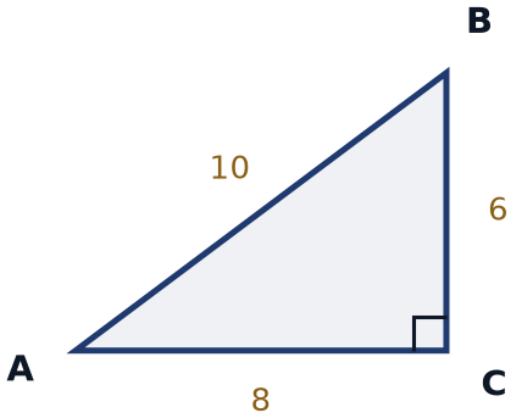
11 [Intermediate] • Target: Degrees, radians, and angles as rotations

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

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12 [Intermediate] • Target: Trigonometric functions in the coordinate plane

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(2)$.



13 [Intermediate] • Target: Trigonometric functions in the coordinate plane

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

14 [Intermediate] • Target: Trigonometric functions in the coordinate plane

A data set recorded for trigonometric functions in the coordinate plane consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

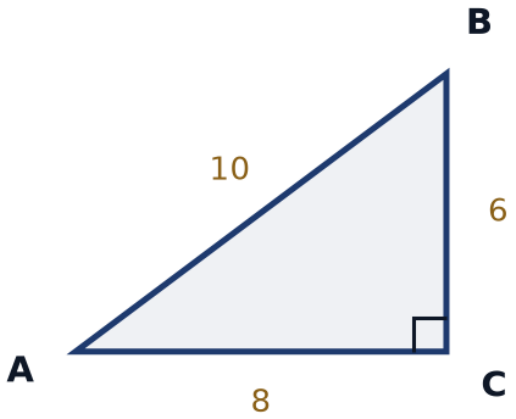
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 2.01 Quadratics & Complex Numbers

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(3)$.

16 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 3.01 Polynomial Functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



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17 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 4.01 Rational Expressions

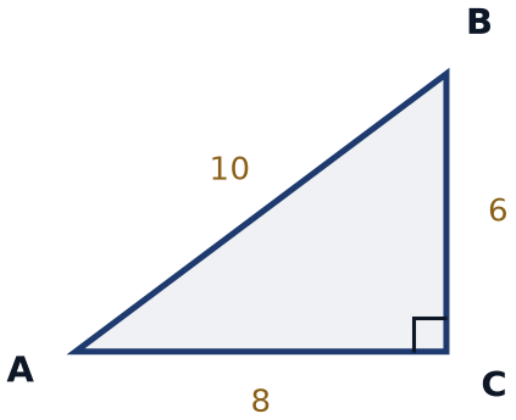
For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(4)$.

18 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

19 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 6.01 Radicals & Rational Exponents

For the model $f(x) = 3x + 3$ in trigonometric functions in the coordinate plane, evaluate $f(5)$.



20 [Challenging] • Target: Trigonometric functions in the coordinate plane • Cross-Lesson: 7.01 Sequences & Series

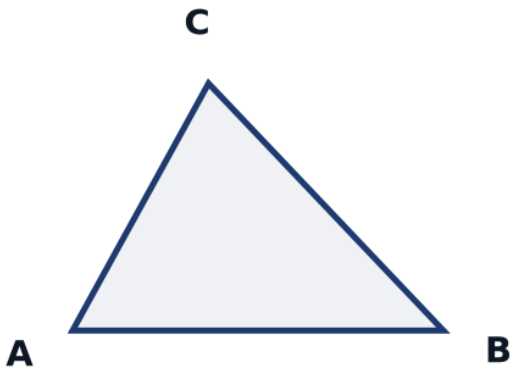
Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 102°

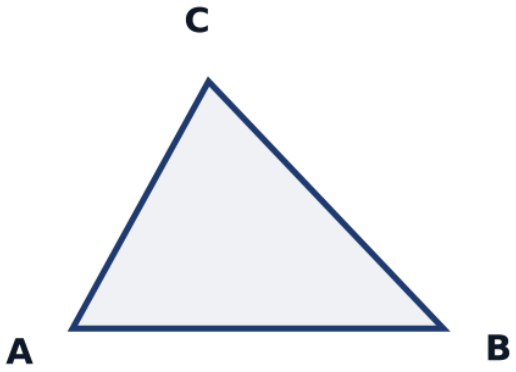
b. 112°

c. 94°

d. 147°

22 [Foundational] • Target: Trigonometric functions as right triangle ratios

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?



a. 18 cm

b. 21 cm

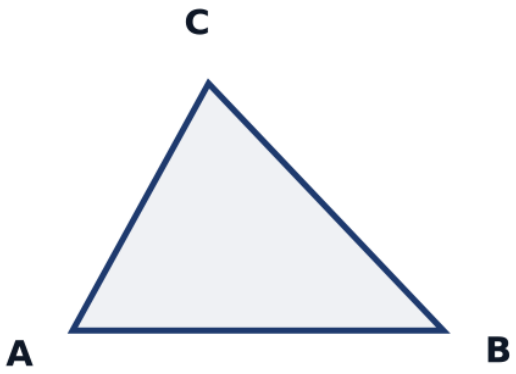
c. 28 cm

d. 63 cm

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23 [Foundational] • Target: Trigonometric functions as right triangle ratios

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?



a. 13

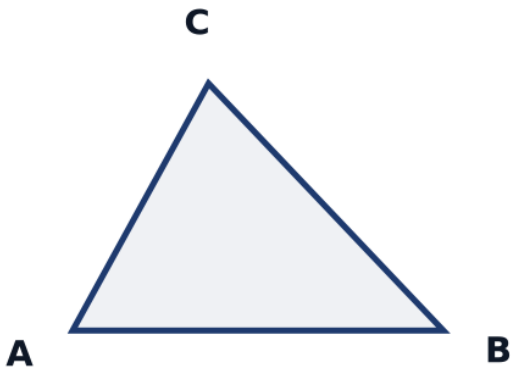
b. 18

c. 16

d. 19

24 [Foundational] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 81°

b. 91°

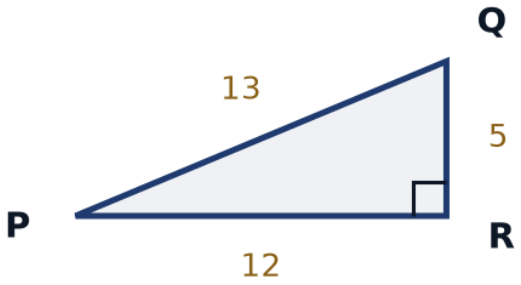
c. 73°

d. 147°

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25 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 101°

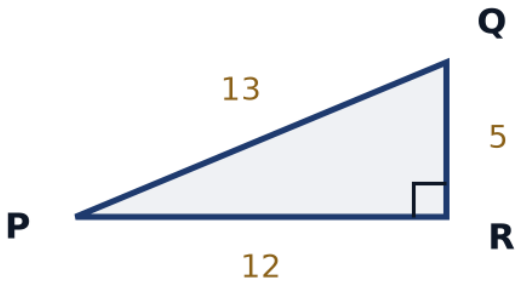
b. 111°

c. 93°

d. 146°

26 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?



a. 21 cm

b. 24 cm

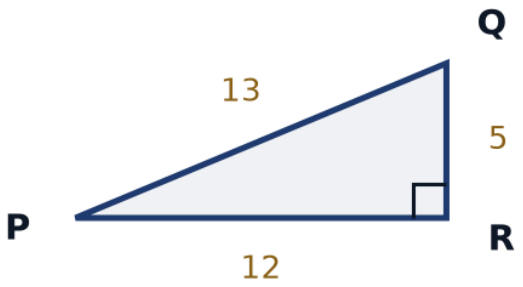
c. 32 cm

d. 72 cm

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27 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?



a. 16

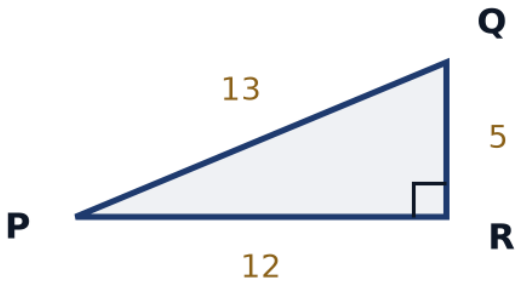
b. 22

c. 20

d. 24

28 [Foundational] • Target: Special right triangles and exact value trigonometric ratios

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 80°

b. 90°

c. 72°

d. 146°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Trigonometric functions as right triangle ratios

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

30 [Intermediate] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95°

b. 105°

c. 87°

d. 147°

31 [Intermediate] • Target: Trigonometric functions as right triangle ratios

In ΔABC studying trigonometric functions as right triangle ratios, $m\angle A = 33^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 88°

b. 98°

c. 80°

d. 147°

32 [Intermediate] • Target: Trigonometric functions as right triangle ratios

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

33 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

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34 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

In ΔABC studying special right triangles and exact value trigonometric ratios, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 94°

b. 104°

c. 86°

d. 146°

35 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

36 [Intermediate] • Target: Special right triangles and exact value trigonometric ratios

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24

d. 28

37 [Intermediate] • Target: Degrees, radians, and angles as rotations

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 16

c. 14

d. 19

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Degrees, radians, and angles as rotations • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 14

b. 21

c. 19

d. 24

39 [Challenging] • Target: Degrees, radians, and angles as rotations • Cross-Lesson: 6.01 Radicals & Rational Exponents

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

40 [Challenging] • Target: Degrees, radians, and angles as rotations • Cross-Lesson: 7.01 Sequences & Series

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 24

b. 31

c. 29

d. 34

41 [Challenging] • Target: Trigonometric functions on the unit circle • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for trigonometric functions on the unit circle consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

42 [Challenging] • Target: Trigonometric functions on the unit circle • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 4x + 4$ in trigonometric functions on the unit circle, evaluate $f(4)$.

a. 20

b. 24

c. 16

d. 24

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43 [Challenging] • Target: Trigonometric functions on the unit circle • Cross-Lesson: 10.01 Matrices & Systems

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

44 [Challenging] • Target: Evaluating trigonometric functions: exact values • Cross-Lesson: 11.01 Probability & Statistics

A data set recorded for evaluating trigonometric functions: exact values consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

45 [Challenging] • Target: Evaluating trigonometric functions: exact values • Cross-Lesson: 2.01 Quadratics & Complex Numbers

For the model $f(x) = 5x + 5$ in evaluating trigonometric functions: exact values, evaluate $f(4)$.

a. 25

b. 30

c. 20

d. 29

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